Exp.no : 08 **LINEAR REGRESSION**

AIM:

Experiment to understand Linear Regression for a given data set.

PYTHON CODE:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df.dropna(inplace=True)
df.info()
df.describe()
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
model.score(x_train,y_train)
model.score(x_test,y_test)
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
In [27]:
model=pickle.load(open('SalaryPred.model','rb'))
In [28]:
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary)
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
# Column Non-Null Count Dtype
--- ------ -------------- -----
0 YearsExperience 30 non-null float64
1 Salary 30 non-null int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

	Out[5]: YearsExperience Salary	
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
Max	10.500000	122391.000000

LinearRegression

LinearRegression()
0.9603182547438908
0.9184170849214232
Enter Years of Experience: 44
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:

RESULT:

The code is executed successfully.